

Appendix H

Smoothed dissipative particle dynamics

H.1 Navier-Stokes equation and Fourier's law

We briefly summarize the most important relations that underlie the [Smoothed Dissipative Particle Dynamics \(SDPD\)](#) method of Español and Revenga [694]. A description of [SDPD](#) fits in this book on particle-based simulations because it provides a particle-based picture of transport in fluids that is compatible with non-equilibrium thermodynamics [57] in that it correctly accounts for entropy production. Moreover, like [DPD](#), it correctly accounts for thermal fluctuations.

As indicated in the main text, the method, which is an extension of the smoothed-particle hydrodynamics method of the 1970s [695–697], aims to represent a continuous hydrodynamic flow field by a set of particle trajectories following equations of motion that, on a large enough scale, mimic the Navier-Stokes equations of viscous flow, combined with those for heat conduction.

The energy of a fluid “particle” i is related to the basic thermodynamic parameters through an equation of state of the form $E_i = E(m, S_i, V_i)$, which remains to be specified. Every particle has a temperature T_i and a pressure P_i given by:

$$T_i = \left(\frac{\partial E}{\partial S_i} \right)$$

and

$$P_i = - \left(\frac{\partial E}{\partial V_i} \right).$$

In the absence of thermal fluctuations, macroscopic transport is described by the Navier-Stokes equation:

$$m\rho \frac{d\mathbf{v}}{dt} = -\nabla P + \eta \nabla^2 \mathbf{v} + (\zeta + \eta/3) \nabla \nabla \cdot \mathbf{v}, \quad (\text{H.1.1})$$

complemented with the (simplified) form of the expression for entropy production due to viscous flow and heat transport

$$T\rho \frac{ds}{dt} = \phi + \kappa \nabla^2 T \quad (\text{H.1.2})$$

where

$$\phi = 2\eta \overline{\nabla \mathbf{v}} : \overline{\nabla \mathbf{v}} + \zeta (\nabla \cdot \mathbf{v})^2$$

and finally, the mass-conservation law

$$\frac{d\rho}{dt} = -\rho \nabla \cdot \mathbf{v} . \quad (\text{H.1.3})$$

In the above equations, ρ stands for the number density: to be consistent with the notation elsewhere in this book, we use a slightly different notation from ref. [694]. $\mathbf{v}$ denotes the flow velocity, P the hydrostatic pressure, η the shear viscosity, ζ the bulk viscosity, s the entropy per particle and κ the thermal conductivity.

H.2 Discretized SDPD equations

The discretized form of Eqs. (H.1.1) through (H.1.3) is:

$$m \dot{\mathbf{v}}_i = -\frac{(\nabla P)_i}{\rho_i} + \frac{\eta}{\rho_i} (\nabla^2 \mathbf{v})_i + \frac{(\zeta + \eta/3)}{\rho_i} (\nabla \nabla \cdot \mathbf{v})_i , \quad (\text{H.2.1})$$

$$T_i \dot{S}_i = \frac{\phi + \kappa (\nabla^2 T)_i}{\rho_i} \quad (\text{H.2.2})$$

and

$$\dot{\rho}_i = -\rho_i (\nabla \cdot \mathbf{v})_i , \quad (\text{H.2.3})$$

where we now use the local expression for density, pressure velocity gradients, and entropy. However, to link the continuum version of the Navier-Stokes equation, we now must give a prescription for computing gradient terms in the vicinity of particle i .

To make this link between the Navier-Stokes equation and the particle-based description, Español and Revenga postulated the following relation between the volume V_i of a particle i to the local density ρ_i around that particle:

$$V_i \equiv 1/\rho_i$$

with

$$\rho_i \equiv \int d\mathbf{r} \rho(\mathbf{r}) W(\mathbf{r} - \mathbf{r}_i) ,$$

where $\rho(\mathbf{r}) \equiv \sum_j \delta(\mathbf{r} - \mathbf{r}_j)$ and $W(\mathbf{r} - \mathbf{r}_i)$ is a normalized weight function:

$$\int d\mathbf{r} W(\mathbf{r} - \mathbf{r}_i) = 1 .$$

Note that the density around particle i includes i itself: hence $\rho_i > 0$. Various choices for the functions form or $W(r)$ are possible; it is convenient to choose a

form for W that has a finite support, i.e., for which $W(r) = 0$ for $r > h$, where h remains to be chosen. Ref. [694] used:

$$W(r) = \frac{105}{16\pi h^3} \left(1 + 3\frac{r}{h}\right) \left(1 - \frac{r}{h}\right)^3 \quad \text{for } r \leq h$$

$$= 0 \quad \text{otherwise .}$$

In what follows, it is not $W(r)$ itself that enters into the equations, but a quantity $F(r)$ defined through the relation:

$$\nabla W(r) = -\mathbf{r}F(r) .$$

For the choice of ref. [694],

$$F(r) = \frac{315}{4\pi h^5} (1 - r/h)^2 .$$

For $F(r_i - r_j)$ we will use the shorthand notation F_{ij} , and similarly $T_{ij} \equiv T_i - T_j$, $\mathbf{r}_{ij} \equiv (\mathbf{r}_i - \mathbf{r}_j)$, $\mathbf{v}_{ij} \equiv (\mathbf{v}_i - \mathbf{v}_j)$ and $\hat{\mathbf{e}}_{ij} \equiv \mathbf{r}_{ij}/|\mathbf{r}_{ij}|$.

It then follows (after some slightly tedious algebra —see ref. [694]) that

$$m\dot{\mathbf{v}}_i = \sum_j \left[\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} \right] F_{ij} \mathbf{r}_{ij} - \left(\frac{5\eta}{3} - \zeta \right) \sum_j \frac{F_{ij}}{\rho_i \rho_j} \mathbf{v}_{ij}$$

$$- 5 \left(\frac{\eta}{3} + \zeta \right) \sum_j \frac{F_{ij}}{\rho_i \rho_j} \hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij} \cdot \mathbf{v}_{ij} . \quad (\text{H.2.4})$$

Eq. (H.2.4), together with

$$\mathbf{r}_i = \mathbf{v}_i ,$$

provides an explicit expression for the time evolution of the positions of the fluid “particles.” Importantly, the equation satisfies Newton’s third law.

We still need to give the discretized form of the entropy production:

$$T_i \dot{S}_i = (\phi)_i - 2\kappa \sum_j \frac{F_{ij}}{\rho_i \rho_j} T_{ij} , \quad (\text{H.2.5})$$

where

$$(\phi)_i = \frac{1}{2} \left[\left(\frac{5\eta}{3} - \zeta \right) \sum_j \frac{F_{ij}}{\rho_i \rho_j} \mathbf{v}_{ij}^2 + 5 \left(\frac{\eta}{3} + \zeta \right) \sum_j \frac{F_{ij}}{\rho_i \rho_j} (\hat{\mathbf{e}}_{ij} \cdot \mathbf{v}_{ij})^2 \right] .$$

To solve these equations, we need an equation of state to relate the pressure to the density at each position: this is our relation $E_i = E(m, S_i, V_i)$, which we can choose to reproduce the properties of the fluid under consideration.

The only thing that remains is to add the noise term, to account for thermal fluctuations in the fluid. This involves adding random noise of the correct amplitude to the entropy of the particles and their velocities in such a way that energy conservation and momentum conservation are satisfied. For details, we refer the reader to ref. [694].